Diagnose, repair, recheck

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify a reasoning error and demonstrate the repaired method on a new problem.

SUCCESS CRITERIA - I can...

1. Locate the first incorrect step.
2. Show a corrected method with units.
3. Apply the repair to a different problem.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “revision” to explain your model or result. Sentence starters are optional.

evidence: observations or measurements that support an idea

revision: a change made after checking evidence or reasoning

assumption: a condition accepted for using a model

01 Warm-up: bring back an earlier idea

A 3 kg block slides down a 30° ramp with 8.7 N friction up the ramp.
Use g=9.8 m/s².
Find along-ramp acceleration and normal force.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$.
For a constant-acceleration interval. Check signs, units and assumptions; keep the original step visible when revising.

Worked reasoning: Claim: a car moving east must accelerate east. Counterexample: it moves east while braking, so acceleration points west. A single valid counterexample can disprove a universal claim.

Your turn: Draw velocity and acceleration arrows for that counterexample. Write a corrected statement about their directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Car braking while moving east

East is right. →

Velocity:

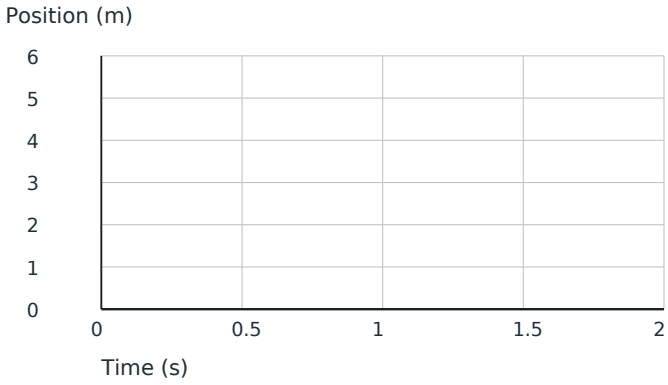
Acceleration / force:

Marzano check - circle: ① ② ③

03 Find the first bad step

A robot starts at x = 5 m and reaches x = 1 m after 2 s. A student calculates v = 1/2 = 0.5 m/s.
Sketch the motion and correct the calculation. Under what condition would dividing final position by elapsed time give average velocity?

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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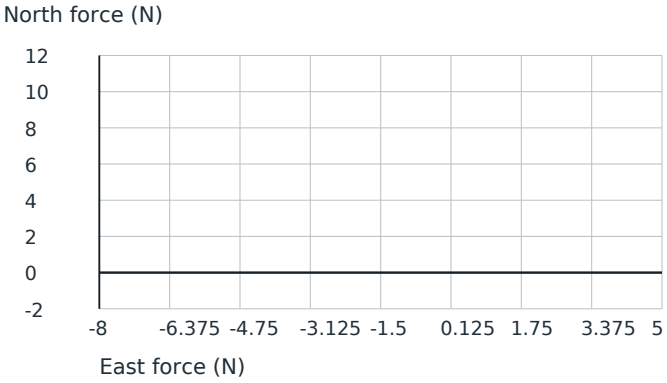
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Check the axis

A rope exerts 10 N at 30° west of north. A student writes $F_x = 10 \cos 30°$ and $F_y = 10 \sin 30°$. East is +x; north is +y.
Sketch the vector and correct both the signs and trigonometric choices.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

05 Fresh recheck

Check your corrected methods on two new cases.
(a) Find the components of an 8 N force at 30° south of east.
(b) A 5 kg cart has 7 N right and 22 N left.
Find its acceleration. Does that alone establish its direction of motion?

Sentence starter: I found … [unit]. This value means … in this situation.



Case (b): mass 5 kg.

Marzano check - circle: 1 2 3

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Continue down each column.

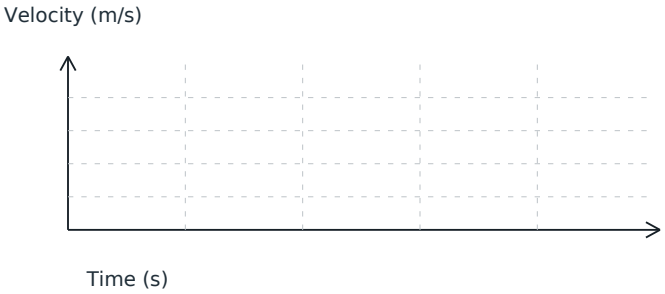
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Velocity changes from -5 to -35 m/s in 5 s, then stays at -35 m/s for 9 s.
Find acceleration and describe the speed trend in each interval.
Sketch the velocity-time model; label the given values.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

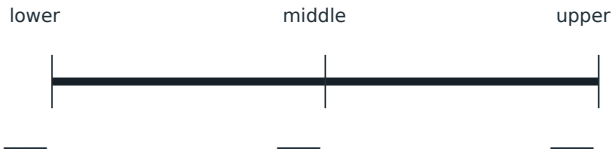


Marzano check - circle: 1 2 3

07 Exit check

A robot starts at 3 ± 1 m and moves at a constant 4 ± 0.5 m/s for exactly 2 s. Treat the input bounds independently.
Find the central final position and full interval.
Name one specific error you checked for.

Sentence starter: The step I would change is ... . The evidence is ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A student calculates acceleration using the applied force alone and gets 5 m/s² for a 2 kg cart. The measured acceleration is 3 m/s². Infer a possible opposing force, then name another explanation to test.

Word help: infer = use evidence to reach a conclusion

Sentence starter: My design shows ... . It meets the requirement by ... .



2 kg cart: measured acceleration 3 m/s² right.

Marzano check - circle: 1 2 3